

WASP-43b

R-2226

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-43_150309 · created 2026-08-15T00:37:19+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2457090.74039 +/- 0.00071 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.197 +/- 0.019
Transit depth (Rp/Rs)^2	3.9 +/- 0.76 [%]
Orbital Inclination (inc)	80.18 +/- 2.07
Airmass coefficient 1 (a1)	1.167 +/- 0.014
Airmass coefficient 2 (a2)	-0.109 +/- 0.015
Scatter in the residuals of the lightcurve fit is	1.2 %
Best Comparison Star	None
Optimal Aperture	4.19
Optimal Annulus	10.55
Transit Duration (day)	0.045 +/- 0.011

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.197 ± 0.019 vs 0.1594 (deviation 1.53σ)
Duration fit vs published	0.045 d vs 0.0512 d (deviation 0.44σ)
Mid-time O–C	2.1 min
Depth SNR	8.0
Residual scatter	1.2 %

Light curve

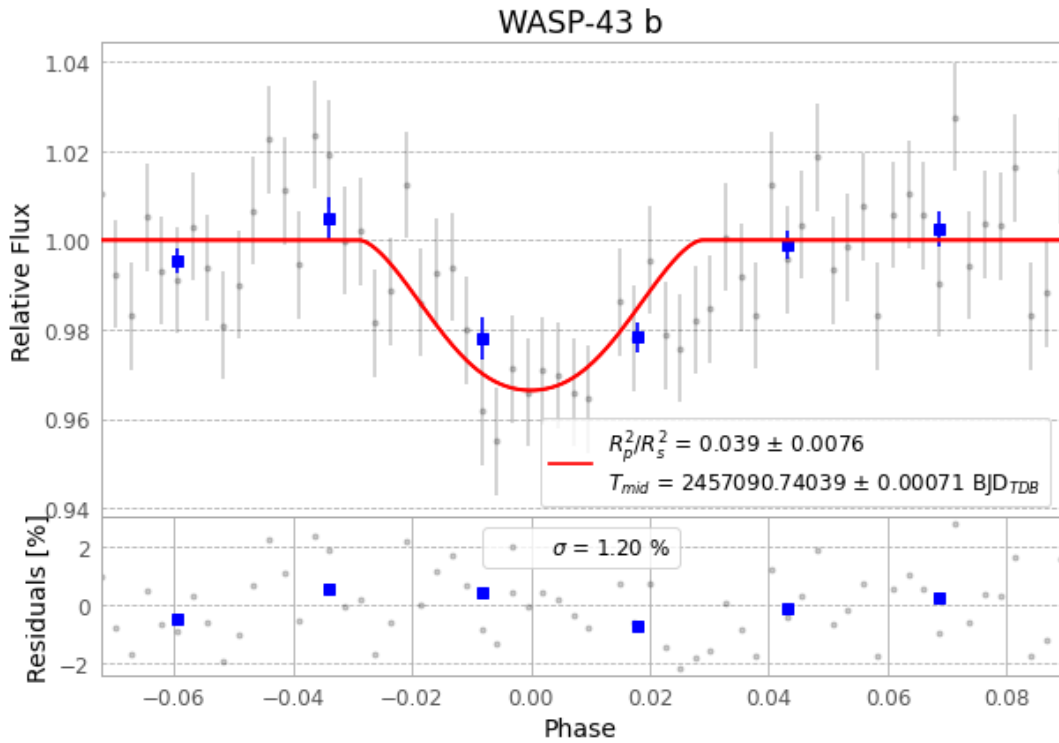


Figure 1 — FinalLightCurve_WASP-43 b_2015-03-08.png (SHA-256 352250891358a3f2...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [156, 213], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 9c3ea91256cfa4fa...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

64 FITS frames (42 MB), WASP-43150309041212.FITS → WASP-43150309072112.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-150309.log (d8ce891ea2c9...), FinalLightCurve_WASP-43 b_2015-03-08.png (352250891358...), FinalLightCurve_WASP-43 b_2015-03-08.csv (bd1c050591f7...)

Compute resources

Wall time 102.9 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-15 00:37Z.